Thermal Impedance Calculation: Aluminium Plate with Liquid Cooling

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1 Introduction and Physical Model

The thermal impedance Z_{th} (or, in steady-state conditions, the thermal resistance R_{th}) quantifies the temperature rise per unit of dissipated power along a thermal path. For a linear, steady-state system:

$$\Delta T = R_{\text{th}}\, P \qquad [\text{K/W}] \tag{1}$$

where ΔT is the temperature difference between the heat source and the coolant reference, and P is the dissipated power in watts.

The system under analysis consists of:

1. an **aluminium plate** with dimensions $40\,\text{mm} \times 160\,\text{mm} \times 5\,\text{mm}$ (the $5\,\text{mm}$ thickness is along the vertical heat-flow axis);

2. a **30 % ethylene-glycol/water mixture** flowing along the lower face of the plate, with a volumetric flow rate $\dot{V} = 24\,\mathrm{l/min}$ through a rectangular channel of height $h_c = 12\,\mathrm{mm}$ and width $w_c = 40\,\mathrm{mm}$.

Heat flows vertically from the top surface (source) to the coolant (bottom). The total thermal resistance is the series combination of two contributions:

$$\boxed{R_{\mathrm{th,tot}} = R_{\mathrm{cond}} + R_{\mathrm{conv}}} \tag{2}$$

The electro-thermal analogy is illustrated in Figure 1: temperature plays the role of voltage, power is the current source, and thermal resistance is the electrical resistance.

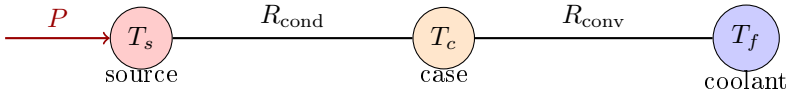


Figure 1: Equivalent electro-thermal circuit. T_s = source temperature, T_c = temperature at the bottom face of the aluminium plate, T_f = bulk coolant temperature.

2 Geometry and Input Data

Table 1: Plate and channel geometry.

Symbol	Quantity	Value	Unit
L	Plate length (flow direction)	160	mm
W	Plate width	40	mm
t	Plate thickness (heat-flow axis)	5	mm
A	Active surface area ($L \times W$)	64	cm^2
h_c	Channel height	12	mm
w_c	Channel width	40	mm
$\dot{V}$	Volumetric flow rate	24	l/min

The active surface area:

$$A = L \times W = 0.160\,\mathrm{m} \times 0.040\,\mathrm{m} = 6.4 \times 10^{-3}\,\mathrm{m}^2 \tag{3}$$

3 Conduction Thermal Resistance

Steady-state heat conduction through a plane wall is governed by Fourier's law:

$$q = -k\frac{\mathrm{d}T}{\mathrm{d}z} \tag{4}$$

where $q\,[\mathrm{W/m^2}]$ is the heat flux and $k\,[\mathrm{W/(m \cdot K)}]$ is the thermal conductivity. For a flat wall of thickness t, area A, and total power $P = q \cdot A$, the conduction thermal resistance is:

$$\boxed{R_{\mathrm{cond}} = \frac{t}{k\,A}} \qquad [\mathrm{K/W}] \tag{5}$$

For general-purpose aluminium (6xxx series): $k_{\mathrm{Al}} = 160\,\mathrm{W/(m \cdot K)}$.

$$R_{\mathrm{cond}} = \frac{5 \times 10^{-3}\,\mathrm{m}}{160\,\mathrm{W/(m \cdot K)} \times 6.4 \times 10^{-3}\,\mathrm{m}^2} = 4.88 \times 10^{-3}\,\mathrm{K/W} \approx 4.88\,\mathrm{mK/W} \tag{6}$$

Note: this value is small compared to the convective resistance, confirming that the aluminium plate is not the thermal bottleneck in this geometry.

4 Convection Thermal Resistance

4.1 Fluid Properties

The coolant is a 30 % ethylene-glycol/water mixture. Thermo-physical properties at a representative operating temperature of $\approx 50\,^{\circ}\mathrm{C}$ are listed in Table 2.

Table 2: Thermo-physical properties of EG 30 % (vol.) at $T \approx 50\,^{\circ}\mathrm{C}$.

Symbol	Property	Value	Unit
ρ	Density	1040	$\mathrm{kg/m^3}$
c_p	Specific heat	3800	$\mathrm{J/(kg \cdot K)}$
μ	Dynamic viscosity	0.82	$\mathrm{mPa \cdot s}$
k_f	Thermal conductivity	0.45	$\mathrm{W/(m \cdot K)}$
Pr	Prandtl number	6.92	–

The Prandtl number:

$$\mathrm{Pr} = \frac{\mu\, c_p}{k_f} = \frac{8.2 \times 10^{-4}\,\mathrm{Pa \cdot s} \times 3800\,\mathrm{J/(kg \cdot K)}}{0.45\,\mathrm{W/(m \cdot K)}} \approx 6.92 \tag{7}$$

4.2 Flow Rate and Mean Velocity

The volumetric flow rate:

$$\dot{V} = 24\,\mathrm{l/min} = \frac{24}{60 \times 1000}\,\mathrm{m^3/s} = 4.0 \times 10^{-4}\,\mathrm{m^3/s} \tag{8}$$

Cross-sectional area of the channel:

$$A_c = h_c \times w_c = 12 \times 10^{-3}\,\mathrm{m} \times 40 \times 10^{-3}\,\mathrm{m} = 4.8 \times 10^{-4}\,\mathrm{m^2} \tag{9}$$

Mean fluid velocity:

$$\bar{v} = \frac{\dot{V}}{A_c} = \frac{4.0 \times 10^{-4}\,\mathrm{m^3/s}}{4.8 \times 10^{-4}\,\mathrm{m^2}} \approx 0.833\,\mathrm{m/s} \tag{10}$$

4.3 Hydraulic Diameter

For a rectangular duct with height h_c and width w_c:

$$D_h = \frac{4\,A_c}{P_w} = \frac{4\,h_c\,w_c}{2\,(h_c + w_c)} = \frac{4 \times 12 \times 10^{-3}\,\mathrm{m} \times 40 \times 10^{-3}\,\mathrm{m}}{2\,(12 \times 10^{-3}\,\mathrm{m} + 40 \times 10^{-3}\,\mathrm{m})} \approx 18.46\,\mathrm{mm} \tag{11}$$

where P_w is the wetted perimeter.

4.4 Reynolds Number

$$\mathrm{Re} = \frac{\rho\,\bar{v}\,D_h}{\mu} = \frac{1040 \times 0.833 \times 18.46 \times 10^{-3}}{8.2 \times 10^{-4}} \approx 19\,512 \tag{12}$$

Since $\mathrm{Re} \gg 4000$, the flow is in the **fully turbulent regime**.

4.5 Nusselt Number Correlations

4.5.1 Dittus–Boelter Correlation

For a heated fluid ($n = 0.4$), valid for $\mathrm{Re} > 10^4$, $0.7 < \mathrm{Pr} < 160$, $L/D_h > 60$:

$$\mathrm{Nu}_{\mathrm{DB}} = 0.023\,\mathrm{Re}^{0.8}\,\mathrm{Pr}^{0.4} = 0.023 \times 19512^{0.8} \times 6.92^{0.4} \approx 132.0 \tag{13}$$

4.5.2 Gnielinski Correlation (Recommended)

More accurate over $3000 < \mathrm{Re} < 5 \times 10^6$, $0.5 < \mathrm{Pr} < 2000$, using the Petukhov friction factor f:

$$f = \big(0.790 \ln \mathrm{Re} - 1.64\big)^{-2} = \big(0.790 \times \ln 19512 - 1.64\big)^{-2} \approx 0.02632 \tag{14}$$

$$\mathrm{Nu}_{\mathrm{G}} = \frac{(f/8)\,(\mathrm{Re} - 1000)\,\mathrm{Pr}}{1 + 12.7\sqrt{f/8}\,(\mathrm{Pr}^{2/3} - 1)} = \frac{(0.02632/8) \times 18512 \times 6.92}{1 + 12.7\sqrt{0.02632/8}\,(6.92^{2/3} - 1)} \approx 144.5 \tag{15}$$

4.6 Convective Coefficient and Resistance

Using the Gnielinski correlation:

$$h_{\mathrm{conv}} = \frac{\mathrm{Nu}_{\mathrm{G}}\, k_f}{D_h} = \frac{144.5 \times 0.45\,\mathrm{W/(m \cdot K)}}{18.46 \times 10^{-3}\,\mathrm{m}} \approx 3523\,\mathrm{W/(m^2{\cdot}K)} \tag{16}$$

Convective thermal resistance over the full plate area A:

$$\boxed{R_{\mathrm{conv}} = \frac{1}{h_{\mathrm{conv}}\,A} = \frac{1}{3523\,\mathrm{W/(m^2{\cdot}K)} \times 6.4 \times 10^{-3}\,\mathrm{m^2}} \approx 44.36\,\mathrm{mK/W}} \tag{17}$$

5 Total Thermal Resistance and Results

$$R_{\mathrm{th,tot}} = R_{\mathrm{cond}} + R_{\mathrm{conv}} = 4.88\,\mathrm{mK/W} + 44.36\,\mathrm{mK/W} \approx 49.24\,\mathrm{mK/W} \tag{18}$$

Table 3: Summary of thermal resistance contributions ($h_c = 12\,\mathrm{mm}$).

Contribution	Expression	Value [mK/W]	% of total
Aluminium conduction	$t/(k_{\mathrm{Al}}\,A)$	4.88	9.9%
Convection (Gnielinski)	$1/(h_{\mathrm{conv}}\,A)$	44.36	90.1%
Total	$R_{\mathrm{cond}} + R_{\mathrm{conv}}$	**49.24**	100%

The source-to-coolant temperature rise for dissipated power P:

$$\Delta T = R_{\mathrm{th,tot}} \times P \approx 49.24 \times 10^{-3}\,\mathrm{K/W} \times P \tag{19}$$

For example, at $P = 1000\,\mathrm{W}$:

$$\Delta T \approx 49.24 \times 10^{-3}\,\mathrm{K/W} \times 1000\,\mathrm{W} \approx 49.2\,\mathrm{K} \tag{20}$$

Comparison with $h_c = 5\,\mathrm{mm}$: increasing the channel height from 5 mm to 12 mm reduces the mean velocity from 2.0 m/s to 0.83 m/s and raises D_h from 8.89 mm to 18.46 mm. Both effects lower turbulence intensity, causing h_{conv} to drop from $\approx 8300\,\mathrm{W/(m^2K)}$ to $\approx 3500\,\mathrm{W/(m^2K)}$ and $R_{\mathrm{th,tot}}$ to more than double (from 23.7 mK/W to 49.2 mK/W).

6 Dynamic Thermal Impedance (RC Model)

In transient operation, the thermal response is modelled by an equivalent RC network. For a single lumped thermal mass (the aluminium plate):

$$C_{\text{th}} = \rho_{\text{Al}}\, c_{p,\text{Al}}\, V_{\text{Al}} \tag{21}$$

With $\rho_{\text{Al}} = 2700\,\text{kg/m}^3$, $c_{p,\text{Al}} = 900\,\text{J/(kg}\cdot\text{K)}$, $V_{\text{Al}} = L \times W \times t = 3.2 \times 10^{-5}\,\text{m}^3$:

$$C_{\text{th}} = 2700 \times 900 \times 3.2 \times 10^{-5}\,\text{m}^3 \approx 77.8\,\text{J/K} \tag{22}$$

The thermal time constant of the conduction path:

$$\tau = R_{\text{cond}} \times C_{\text{th}} = 4.88 \times 10^{-3}\,\text{K/W} \times 77.8\,\text{J/K} \approx 0.38\,\text{s} \tag{23}$$

The thermal impedance in the frequency domain (single-pole Foster model):

$$Z_{\text{th}}(j\omega) = \frac{R_{\text{th,tot}}}{1 + j\omega\tau} \tag{24}$$

For a power pulse of duration t_p, the peak temperature rise is estimated from the transient impedance curve:

$$\Delta T_{\text{peak}} = Z_{\text{th}}(t_p) \times P_{\text{peak}} \tag{25}$$

7 Model Validity and Assumptions

1. **Steady-state analysis.** The calculation provides the static resistance R_{th}. For transient analysis a full Foster or Cauer RC network is required.
2. **One-dimensional heat flow.** Valid for uniform power distribution. Localised sources (e.g. IGBT chips) introduce an additional spreading resistance.
3. **Smooth-wall channel.** The Gnielinski correlation applies to smooth-walled ducts. Pin-fin or corrugated structures significantly enhance h_{conv}.
4. **Constant fluid properties.** Properties of the glycol mixture vary with temperature; a more accurate analysis would iterate on the mean film temperature.
5. **Contact resistance.** The interfacial resistance between the aluminium plate and the channel wall is not included; it depends on surface roughness and any thermal interface material applied.
6. **Channel height impact.** A larger h_c increases D_h and reduces $\bar{v}$ at constant flow rate, both lowering the Nusselt number and increasing R_{conv}. For optimal performance the channel should be as shallow as mechanically permissible, or equipped with internal fins.

8 Conclusions

The steady-state total thermal resistance of the aluminium-plate/glycol-cooled system with channel height $h_c = 12\,\text{mm}$ is:

$$R_{\text{th,tot}} \approx 49.24\,\text{mK/W} \tag{26}$$

Convection dominates (≈90 % of the total). The 5 mm aluminium layer contributes only 4.88 mK/W and is not the limiting factor. Compared to the shallower $h_c = 5\,\text{mm}$ configuration ($R_{\text{th,tot}} \approx 23.7\,\text{mK/W}$), the deeper channel more than doubles the total thermal resistance. To improve cooling performance one should reduce the channel height, increase the flow rate, or introduce pin-fin structures to enhance convection.